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RESEARCH ARTICLE

Adaptive Personalization of E-Commerce Interfaces Based on User Behavior Data

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ABSTRACT In the dynamic environment of e-commerce, personalization of user interfaces (UI) is crucial for user engagement, as static designs are unable to meet the diverse needs of users. This article presents an innovative, automated framework for adaptive personalization of user interfaces in e-commerce based on user behavior data. The methodology consists of two basic algorithms. First, the contextual algorithm selects the optimal clustering method (evaluating K-means, BIRCH, etc.) by synthesizing performance indicators (e.g., Silhouette Score, Calinski-Harabasz) with business constraints to effectively segment users. Second, the self-adjusting algorithm automatically accepts or rejects ''micro-changes'' to the user interface for each cluster through a sequential A/B testing process, regulated by key performance indicators, including conversion rate (CR), average order value (AOV), and proposed partial conversion rate (PCR). We verified this structure on a real e-commerce platform by analyzing 438,261 user sessions to select K-means as the optimal clustering method. Subsequent tests conducted on 99,193 sessions showed a significant increase in performance: one user cluster achieved a 27.56% increase in CR and a 68.09% increase in AOV, while another saw an 18.87% increase in CR. Theoretically, this work contributes to a closed user-oriented adaptation model that goes beyond product recommendations and encompasses the overall evolution of the user interface. In practice, it provides e-commerce companies with an automated, data-driven tool for dynamically tailoring interfaces to different user segments, leading to significantly higher revenues and greater customer satisfaction.

INDEX TERMS Adaptive system, e-commerce, user interface, personalization.

I. INTRODUCTION

The e-commerce industry has grown exponentially in recent years, with millions of users shopping online across multiple platforms. In this competitive landscape, providing a user-friendly and personalized experience is critical for e-commerce businesses to stand out and thrive. Reports from leading consulting firm McKinsey&Company show that 76% of respondents consider personalization to be a key factor when considering purchasing products from a new brand, with 78% believing that personalized content increases the likelihood of a repeat purchase [1]. Personalized navigation (75%), relevant product/service recommendations (67%) and personalized communication (66%) were cited as the most common expectations for personalization [2]. These needs can be met in various ways, e.g. through high-quality logistics [3], but also by providing an appropriate user interface (UI). The UI plays a critical role in shaping the overall user experience, but designing a static UI that meets the diverse needs and preferences of individual users is a challenging task.

Adapting the user interface of information systems to the needs and behavior of users has been practiced for several decades. A review of different approaches based on articles published between 1960 and 2015, presented by Miraz et al. [4], allows us to trace the evolution of adaptive

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user interface (AUI) approaches over the years. Among the solutions discussed, there are some that relate to web-based interfaces commonly used in e-commerce systems, such as online stores. It is worth noting that web shop customers are different from users of typical business IT systems, such as Enterprise Resource Planning (ERP) systems, Warehouse Management Systems (WMS), Workflow Systems (WF), and so on [5]. First, the group of users is potentially many times larger than the group of employees in a company that wants to personalize the user interface of its ERP system; second, customer behavior in online shops is not structured by intra-organizational business processes and is therefore more difficult to predict. There is therefore limited scope for simply transferring the functioning of AUIs in typical management information systems to e-commerce systems.

However, this does not mean that the user interfaces of e-commerce systems cannot be personalized. It is possible, but it requires an approach that takes into account the specifics of the data on which UI personalization is based, as well as the requirements and technical limitations of such a solution. For many years, by far the most popular approach has been to implement recommendation services, which use Artificial Intelligence (AI) methods to look for patterns in customers' behavior and product features to generate lists of goods that are displayed on e-commerce sites as *Recommended for you*, *Other customers have also bought*, etc. Such solutions have a limited impact on the design of the UI, as they personalize only a small part of the content presented to the viewer. While personalizing content - such as recommending specific products - is now a well-established practice, it predominantly addresses *what* is presented to the user. In this work, we show that a crucial, yet under-explored, dimension is the personalization of the user interface (UI) itself, which governs *how* and *where* that information is displayed. A reliance on a static, one-size-fits-all UI inevitably creates friction for distinct user segments. For instance, a goal-oriented user may be frustrated by an interface optimized for browsing and discovery, while an exploratory user may find a purely functional layout uninteresting. This mismatch between user intent and interface design translates directly into measurable business losses. These include elevated bounce rates as users abandon unintuitive interfaces, diminished conversion rates as certain segments struggle with an inefficient purchase path, and a lower customer lifetime value because the overall experience fails to foster the loyalty required for repeat transactions. By dynamically adapting UI elements like navigation structure, filter presentation, and layout hierarchy, it is possible to mitigate these issues, reduce cognitive load, and more effectively meet the specific needs of diverse user segments. This article, therefore, moves beyond content-level recommendations to propose a framework for adapting the entire interface structure based on user behavior.

The implementation of such a comprehensive approach requires the design of a multi-component system that includes the collection of data on customer behavior, followed by data processing to generate consistent groups of users, for whom a specific UI variant is designed. This challenge has been addressed in the design of the AIM^2 platform, enabling it to serve a multi-variant e-commerce user interface. One of the features implemented is the UI self-adaptation mechanism discussed in this paper. The algorithms adopted in deciding whether to accept or reject potential changes to the user interface are an important element of the solution described. They can give a set of recommendations to a human overseeing the UI customization process, but they can also act in an automated way, reviewing subsequent changes and deciding on whether to implement a particular variant of the UI for a specific group of customers. It is worth adding that this approach is adapted to the concept of re-centering the user's role in recommender systems by involving the user more fully in the feedback loop. Users are no longer just a source of raw data, but are integral in UI adaptation.

Therefore, the main contribution of this article is not simply the application of yet another machine learning technique to personalization. We present a fully automated framework that stands out from existing solutions in three key ways. First, it focuses on holistic user interface adaptation—modifying the layout and interaction mechanisms—rather than limiting personalization to content recommendations. Second, it presents a closed, self-adjusting system that continuously tests, verifies, and implements user interface changes in a real-world environment without manual intervention. Third, it implements a user-centric evolution model in which user behavior becomes the direct driver of iterative interface improvement for different segments. This article describes in detail the algorithms that power this framework and demonstrates their practical impact through real-world experimental research.

The paper is organized as follows. Section II discusses previous work on automatic adaptation of user interfaces and the use of various AI-based methods for clustering e-commerce customers. Section III presents the proposed recommendation algorithms implemented in the AIM^2 platform. The experiments conducted are described and discussed in detail in Section IV. Section V presents limitations and suggests directions for further research. Finally, Section VI summarizes the work and highlights theoretical and practical implications.

II. RELATED WORK

A. THE EVOLUTION OF ADAPTIVE USER INTERFACES

The user interface can improve the user experience (UX) and make a website more interactive, especially for e-commerce applications [6]. The problem of user interface customization using recommendation systems was studied before the widespread use of artificial intelligence-based solutions [7]. Miraz et al. [4] summarized various approaches to web site personalization, citing a number of studies on adaptive hypermedia solutions from 1996-2002. The interest in this topic at a time when web technologies

were just beginning to be widely used in business shows a recognition of the opportunities that a personalized user interface brings to e-commerce. That publication also reviews material on the importance of user characteristics and user customization in the design of the visual layer of various types of information systems, starting with the first applications in the 1960s. Despite the passage of several decades, the topic is still relevant, only the technological aspects have changed, both in terms of UI delivery and processing data on user behavior. Due to the scope of the analyzed publications (up to 2015), relatively few of them explicitly touched on the use of artificial intelligence or machine learning (ML), but the summary ''[For] a machine-learning based algorithm learning from human users, an efficient training vector for fast optimal convergence of the algorithm is a necessity'' well describes the development trends of AUI systems in the following years. Today, an increasing number of solutions for analyzing the behavior of users of IT systems (including e-commerce [8]) use AI/ML to detect dependencies and relationships that are invisible to traditional methods based on, for example, decision rules. However, concepts are now emerging for the advanced use of machine learning to segment e-commerce customers for advanced UI personalization [9]. In summary, it can be concluded that interaction is a very important element in the development of modern IT systems, forming an integral part of the Industry 5.0 paradigm [10].

B. CONTENT-CENTRIC PERSONALIZATION

The architecture of such a solution, which collects data on customer behavior in an online store, analyzes them and generates information that enables personalization, can be similar to the architecture of recommendation systems used in other business areas [11]. Nowadays, various techniques based on artificial intelligence are used to group e-commerce customers on the basis of their behavior and expectations [12]. The choice of approach depends primarily on the objectives of the application of the system [13]. In e-commerce practice, including e-commerce and m-commerce [14], product recommendation systems are most commonly used to present a list of items that might interest a customer [15]. In such an application, CBF-based systems that enable inference based on similarities of both objects and users provide good results [16]. Various approaches based on a wide range of artificial intelligence techniques, such as deep learning and distributed expression [17], fuzzy sets [18], or clustering [19], can be used to analyze e-commerce customers [20]. The latter paper discusses the application of K-means clustering to product recommendations. This algorithm is commonly used for customer segmentation in e-commerce, so it can also be used for methods based on user similarity. Alves Gomes and Meisen [21], based on publications from 2000 to 2022, provided a summary of contemporary segmentation methods used in e-commerce and indicated that clustering algorithms, in particular K-means, are the most popular. From the point of view of providing a dedicated interface to e-commerce customers, the ability to segment them is crucial, and various clustering methods using AI algorithms are commonly used for this purpose [22].

The development of mechanisms for customizing the user interface of a web application has been going on for many years. Much of this has involved the use of recommendation systems [23] and machine learning [24], [25]. A model-driven, ontology-based architecture for generating adaptive user interfaces based on semantic models, including the user context, and ontology models of adaptation rules has been presented in [26]. A computational method that supports the design and evaluation of e-commerce web interfaces by finding the optimal placement of interactive elements is described in [27]. Optimizing the experience associated with live-streaming commerce interfaces is also very important [28]. Moreover, the future may belong to a real-time neuro-adaptive interface for e-commerce tasks [29].

An important issue related to adaptive e-commerce interfaces is performance evaluation, which can include qualitative aspects, such as user acceptance analysis [30], as well as quantitative aspects, such as e-commerce performance indicators. Studies have explored various factors that influence the effectiveness of proposed solutions, such as customer segmentation methods, user profiling, recommendation systems, and user interface design [31].

C. CUSTOMER SEGMENTATION USING CLUSTERING METHODS

From the point of view of designing specific variants of e-commerce user interfaces, one of the key features should be the division of customers into coherent and meaningful groups. Such possibilities are provided by methods for clustering clients [32], [33]. Various algorithms can be used for this purpose. The use of Hierarchical Clustering (HAC) in e-commerce applications is discussed in [34]. The HAC algorithm used for customer segmentation in general has been described in [35] and the results can also be used to segment e-commerce customers. The K-means algorithm can be used for e-commerce customer segmentation to obtain groups of customers with different characteristics [36]. K-medoids, a variant of the K-means method, uses medoids instead of the mean and can be applied for e-commerce customer segmentation [37]. Another method - DBSCAN (Density-Based Spatial Clustering of Applications with Noise) - has a disadvantage that makes it difficult to use for grouping e-commerce customers. Its use leads to the generation of an unpredictable number of clusters, which makes it impossible to prepare variants of the interface for each cluster [38]. In this context, it is important to measure the quality of soft computing tools used in e-commerce [39]. This observation significantly influenced the research concept described below. These issues are particularly important from the point of view of the AIM^2 platform mentioned above,

since the clustering module is a key element in preparing groups of customers for further analysis due to business assumptions.

It is worth noting that additional opportunities to gather information about customer behavior and preferences are now emerging beyond the online store itself. Examples include the increasing use of Internet of Things (IoT) applications, which provide a broader view of the possibilities for generating recommendations in e-commerce [40]. It is possible that such datasets will also be useful in systems that recommend changes to user interfaces, including web-based systems [41].

When analyzing recommendation systems, it is necessary to consider the possibility of suggesting wrong solutions. This can be the result of a number of factors [42]: poor quality input data, poor algorithms, or even intentional misinformation [43]. This means that when proposing new solutions, user-based verification and correction mechanisms should be introduced to reduce the risk of generating ineffective (or even counterproductive) recommendations.

Based on the literature on the subject, our research is grounded in several conclusions. While content personalization is already mature, the literature points to a significant opportunity for holistic adaptation of the user interface structure itself. From a theoretical standpoint, many challenges in UI design can be framed through the lens of cognitive load theory, which posits that a user's working memory is a limited resource. Ineffective UIs impose extraneous cognitive load, hindering task completion and decision-making. The existing literature on e-commerce personalization, particularly content recommenders, sometimes addresses the cognitive load associated with choice overload. However, these approaches often fail to consider the navigational or interactional cognitive load imposed by the interface layout, filtering systems, and information hierarchy. This reveals a critical research gap - the lack of automated frameworks that adapt the holistic UI to manage the user's total cognitive load, rather than just one component of it. Our work is positioned directly within this gap by proposing a system that personalizes the interface structure to create more efficient and less cognitively demanding user journeys.

Moreover, the challenges associated with specific clustering methods highlight the need for a practical, context-aware selection process that balances technical performance with business constraints, rather than relying on a single, pre-selected algorithm. Finally, the risk of implementing ineffective modifications clearly shows that a robust, closed-loop validation mechanism is essential. Any automated system must continuously verify the positive impact of changes based on specific performance indicators. These conclusions directly motivate the two-pillar structure of our proposed architecture - an algorithm for the optimal selection of a clustering method and an algorithm for the automatic validation of micro-changes applied to the user interface based on KPIs.

III. THE METHOD-SELF-ADAPTATION MECHANISM

A. SYSTEM ARCHITECTURE

The preparation and delivery of user interface variants presented to the customers of an online shop requires the design and implementation of a comprehensive system that combines the collection and analysis of data on visitor behavior, the design and delivery of UI modifications, and the management of the process of fine-tuning the UI design to meet the specific needs and requirements of particular user groups. The proposed architecture of such a mechanism, consisting of several components that work together to provide a seamless user experience, is shown in Fig. 1. The feedback loop emphasizes the key role of users in the entire recommendation system.

1) USER DATA LAYER

The procedure responsible for the collection of behavioral data is not dissimilar to the typical collection of information about user activities by web-based systems, especially those implementing so-called clickstream. This type of data is often used in in-depth web analytics to track and predict customer behavior [44] and is structured to include information about the web pages visited, time spent browsing, actions taken and choices made. It is worth noting that one of the major challenges in this part of the self-adaptation process is anonymizing the collected data, while allowing particular UI variants to be served to specific customers and taking into account feedback from external systems (e.g. ERP). One possible solution is to use unique customer identifiers (such as the Universally Unique Identifier - UUID) and store them in cookies on a customer's device. This approach is not without its drawbacks (for example, it does not take into account the possibility of multiple users using the same device), but it is relatively easy to implement and ensures anonymity at the data collection stage. Data privacy issues [45] must also be taken into account when designing this layer, in particular, obtaining the relevant consents and performing the necessary anonymization. However, it should not be forgotten that the more information customers share with an online store operator, the better the personalization they can obtain [46].

2) SEGMENTATION LAYER

Based on the behavioral data collected, customers should be grouped to obtain sets of users with similar characteristics. This is done using the segmentation component, which includes data preprocessing and clustering functions. This part of the system is associated with the first of two algorithms that enable the UI's self-adaptation mechanism to operate automatically - the selection of a clustering method. This feature is important for two reasons. First, there are many clustering algorithms that can be used to segment e-commerce customers. In addition, most of them allow various forms of parameterization, which multiplies the number of potential options from which to choose an approach to user

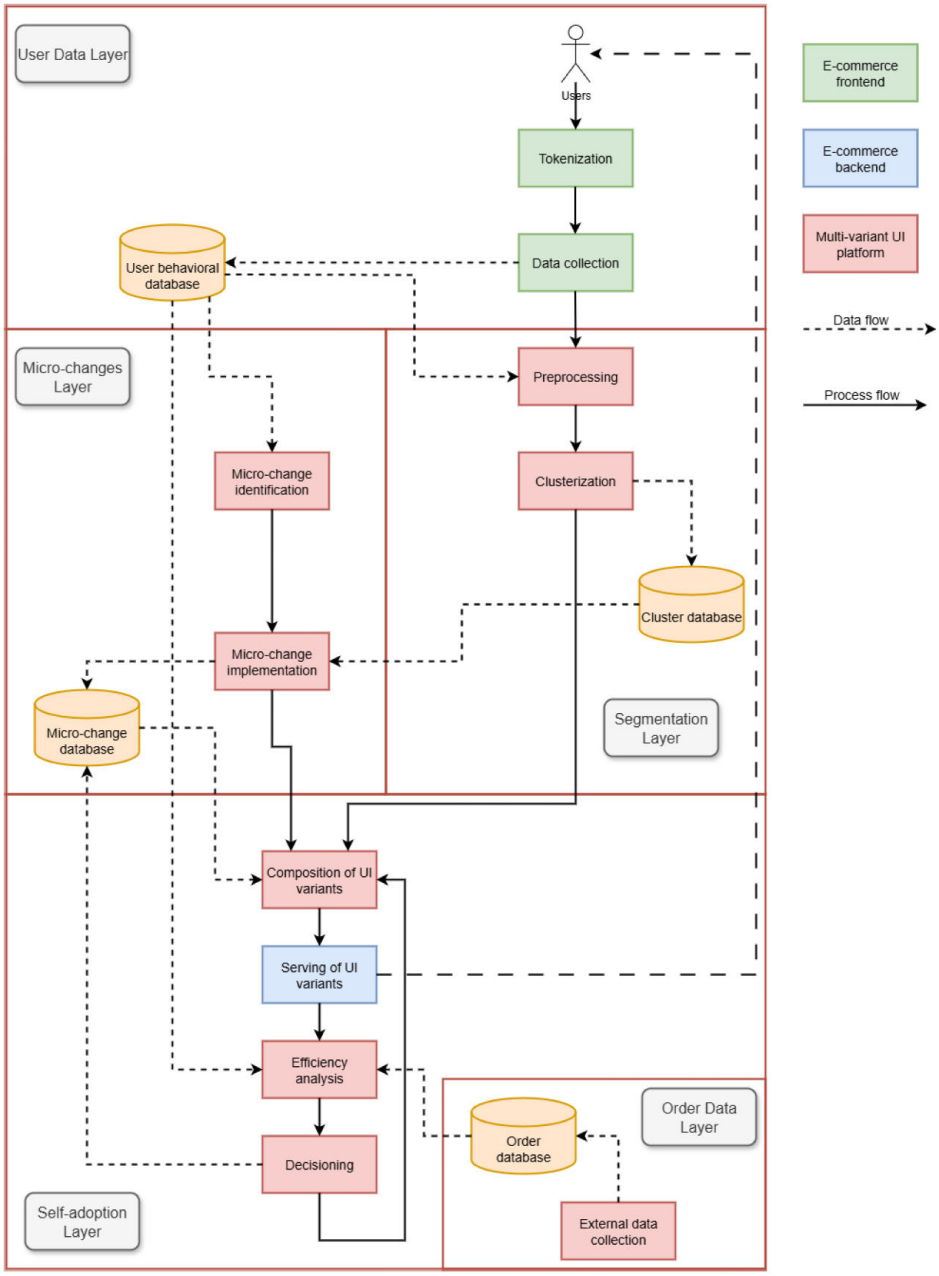


FIGURE 1. System architecture for UI self-adaptation.

segmentation. Second, the quality of clustering depends on the training data, so it is not possible to predict in advance which algorithm with which parameter configuration will work best in a given business context. In the case of a supervised solution, the choice of a clustering method can be made by the administrator, but in the case of an autonomous solution, it is necessary to algorithmize this operation.

3) MICRO-CHANGES LAYER

The next element of the platform concerns the design and implementation of user interface modifications. This is the only part of the proposed system that requires initial human analysis and development. The idea of UI self-adaptation in the proposed solution is based on the concept of a *micro-change*, understood as a small but potentially significant adjustment to the appearance of a page in an online shop. A fundamental issue in micro-change management is to identify the places (areas) on the webshop pages that can and should be modified. Sources of inspiration for UX experts can come from ideas discarded during the phase of designing the standard interface variant and from suggestions resulting from the analysis of data collected on customer behavior, especially after taking into account their clustering. Examples of such changes include modifications to the



FIGURE 2. Examples of micro-changes of filters.

appearance of filters in product listings (Fig. 2). The scope of the designed micro-changes is based on the decisions of an expert. They can be relatively small, such as *micro-change A* and *micro-change B*, which differ in the expansion of some filters, or they can have a greater impact on the appearance of the entire webshop page, such as *micro-change C*, which changes the organization of filters and moves some of them above the products displayed in the list. The second element of the module, which is necessary for the service of micro-changes, is their implementation within the online shop system. For instance, Figure 3 illustrates a typical micro-change applied to a checkout page. Here, Variant B modifies the layout and introduces a 'Top Sellers' recommendation module. This change is not merely aesthetic - it represents a testable hypothesis that displaying relevant products at this critical stage may reduce cart abandonment or increase the Average Order Value. The purpose of this is to make it technically possible to display a specific change to the group of customers to whom it has been assigned, as part of the UI variant assigned to them. Again, the involvement of a human expert (in this case a software developer) is essential.

4) SELF-ADOPTION LAYER

Another component of the proposed solution is the management of UI variants prepared for different customer clusters. A UI variant is a set of micro-changes that optimize the

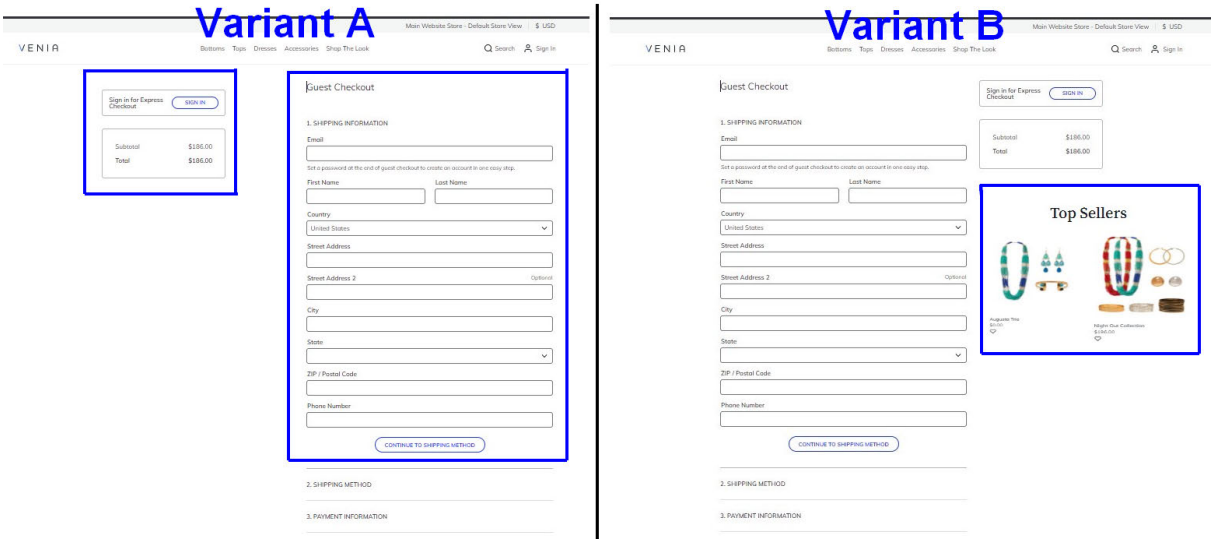


FIGURE 3. Example of a UI micro-change applied to a checkout page.

user experience and are driven by the needs, expectations, behaviors and decisions of a specific group of customers. A variant design process can be either human-driven or the result of a self-adaptation mechanism. In the latter case, UI customization takes place in a feedback loop. It starts with a standard UI variant to which the first micro-change is applied from a prepared list of potential changes. This variant is then deployed to a subset of customers from a cluster for a period of time (measured in days, weeks, or number of user sessions) to gather feedback. At the same time, the remaining members of that cluster are served by the default UI variant, forming a comparison group. The next step is to calculate and analyze selected indicators to determine the impact of the introduced micro-change on customer behavior and decisions. The decision to accept or reject a micro-change is based on a comparison of the values of these indicators for the group of users served with the tested UI variant and for the comparison group. The adaptation of UI variants (testing and acceptance of micro-changes) takes place in parallel and independently for each customer cluster generated. This means that the pace of change of variants can vary according to the activity of the users assigned to specific clusters.

5) ORDER DATA LAYER

The last element of the proposed solution is optional and its possible implementation is aimed at improving the quality of the feedback received. The idea is to include the whole process of orders placed in the online shop within the decision-making process. It is not possible to collect information about the fulfillment and shipping status of an order during the tracking of customer activity, as this is set later, usually in the warehouse system. Therefore, only those orders that have been finally completed, sent to the customer, and not returned within a certain period of time are included in the performance analysis. It is necessary to integrate the data from the external system and calculate the values of the performance indicators linked to the orders on the basis of these data.

Delivering personalized content cannot cause significant delays. To address the critical issues of computational cost and scalability, especially for the real-time serving of UI variants, the architecture incorporates several key design principles. First, a user's cluster assignment is stored server-side and updated only when a new session starts. This avoids costly database lookups for every request and ensures minimal latency. Second, the HTML, CSS, and JS components that constitute each UI variant are pre-rendered and cached. This allows the system to deliver the correct personalized interface with negligible CPU overhead, as it becomes a matter of serving static assets rather than dynamic generation. Finally, the message queue for ingesting behavioral data acts as an asynchronous buffer, decoupling the data collection from the user-facing request-response cycle. This ensures that the data ingestion load does not impact the performance of the online store. These strategies are fundamental to ensuring the system is both scalable to high traffic volumes and practical for real-world implementation.

B. DATA PROTECTION AND LEGAL COMPLIANCE

The processing of user data, even for the purpose of improving user experience, must be carried out in strict compliance with data protection regulations such as the General Data Protection Regulation (GDPR). The proposed framework has been developed in accordance with the following privacy-by-design principles to ensure legal and ethical compliance.

1) PSEUDONYMIZATION AND ANONYMIZATION

The privacy model is based on the immediate pseudonymization of user data. All behavioral analyses are performed

on data linked to an unidentifiable universal identifier (UUID) rather than personal data such as names or email addresses. This ensures that clustering and decision-making algorithms operate on anonymous behavioral patterns rather than identifiable personal data.

2) CONSENT MANAGEMENT

The activation of personalization and data collection mechanisms is strictly dependent on obtaining the user's explicit and informed consent. The proposed architecture is integrated with a consent management platform. Users who do not give their consent or later withdraw it will only have access to the default, non-personalized user interface, and their behavioral data will not be collected or processed by this system.

3) DATA MINIMIZATION

The architecture complies with the principle of data minimization. Only data that is absolutely necessary to customize the user interface is collected. This includes events related to user interaction (e.g., clicks, scrolling, time spent on the page) and anonymous order information. The system is explicitly designed to avoid collecting sensitive or unnecessary personal data.

4) SUPPORT FOR DATA SUBJECTS' RIGHTS

The system is designed to support the rights of data subjects, including the right of access and the right to erasure (''right to be forgotten''). The architecture includes a mechanism whereby a verified user request can result in the deletion of all data relating to behavior associated with their specific UUID from the system's databases. This gives individuals control over their data.

C. ALGORITHM FOR SELECTING A CLUSTERING METHOD

Partitioning an online store's customers based on an analysis of their behavior using clustering algorithms makes it possible to uncover dependencies and relationships that are difficult to discover using traditional approaches, such as decision rules. Selecting and preparameterizing a specific clustering method is undoubtedly a challenge. This is due to the variety of approaches that can be used. An additional factor influencing the choice of clustering algorithm is the lack of ex ante knowledge about cluster shapes, the optimal number of clusters, or the quality of the learning data.

To facilitate this decision and to support its automation when the system operates in autonomous mode, the following algorithm is proposed for recommending the clustering method to be used in the automatic adaptation process of the e-commerce user interface (Fig. 4). It consists of three main phases: initialization, comparison and reasoning.

1) INITIALIZATION

The initialization phase sets the framework for benchmarking the clustering algorithms that may be used. It should primarily take into account the technical issues related to clustering itself, as well as the business context in the application of the results obtained.

The first set of factors is related to the available (implemented) clustering algorithms and their computational complexity, which is a critical consideration for large-scale systems. For instance, while methods like Agglomerative Clustering can be effective for smaller datasets, their typical $O(n^2)$ complexity makes them computationally prohibitive for millions of user sessions. In contrast, algorithms like K-means exhibit linear scalability (approximately O(n)), and methods such as BIRCH were specifically designed for very large datasets by processing data in a single pass. Our selection framework inherently accounts for this by evaluating performance metrics that can include execution time. It is also important to note that this benchmarking is an offline, periodic process performed on a training dataset for model selection, not a real-time computation for every user session, thus managing the computational burden. This design ensures that even for platforms with millions of users, the clustering process remains feasible without impacting the real-time performance of the e-commerce site if the right clustering algorithm is chosen.

The second group is related to the constraints of assigning UIs to web shop customers. Preparing a customized UI variant is a non-zero cost, so it is necessary to determine the minimum number of customers for which such a task is worthwhile. This is also important, because practice shows that only a subset of clustered customers will use the webshop again, so the number of clusters should be appropriate to serving each client via a suitably assigned interface variant.

In the proposed solution, the initialization phase includes the following tasks:

- preparation of a list of clustering algorithms to be analyzed, together with an optional list of parameters (e.g. determining how the initial clusters are generated, the maximum number of iterations) and their possible values,
- specifying the minimum size of a single resultant cluster, expressed in absolute (number of clients in the cluster) or relative (percentage of the total population of clients analyzed) terms,
- defining the maximum number of resultant clusters, according to the number of UI variants to be designed.

The available clustering options should be ranked according to the smallest number of resultant clusters (k), if the algorithm supports such an initial parameter.

2) COMPARISON

The second phase involves iterative clustering of the learning data set using the identified options (including the algorithm and clustering parameters).

This is followed by calculating the values of the metrics that describe the quality of the clustering. Examples of metrics that can be used to evaluate the quality of clustering are the Silhouette Score (SI) [47], Calinski-Harabasz Score

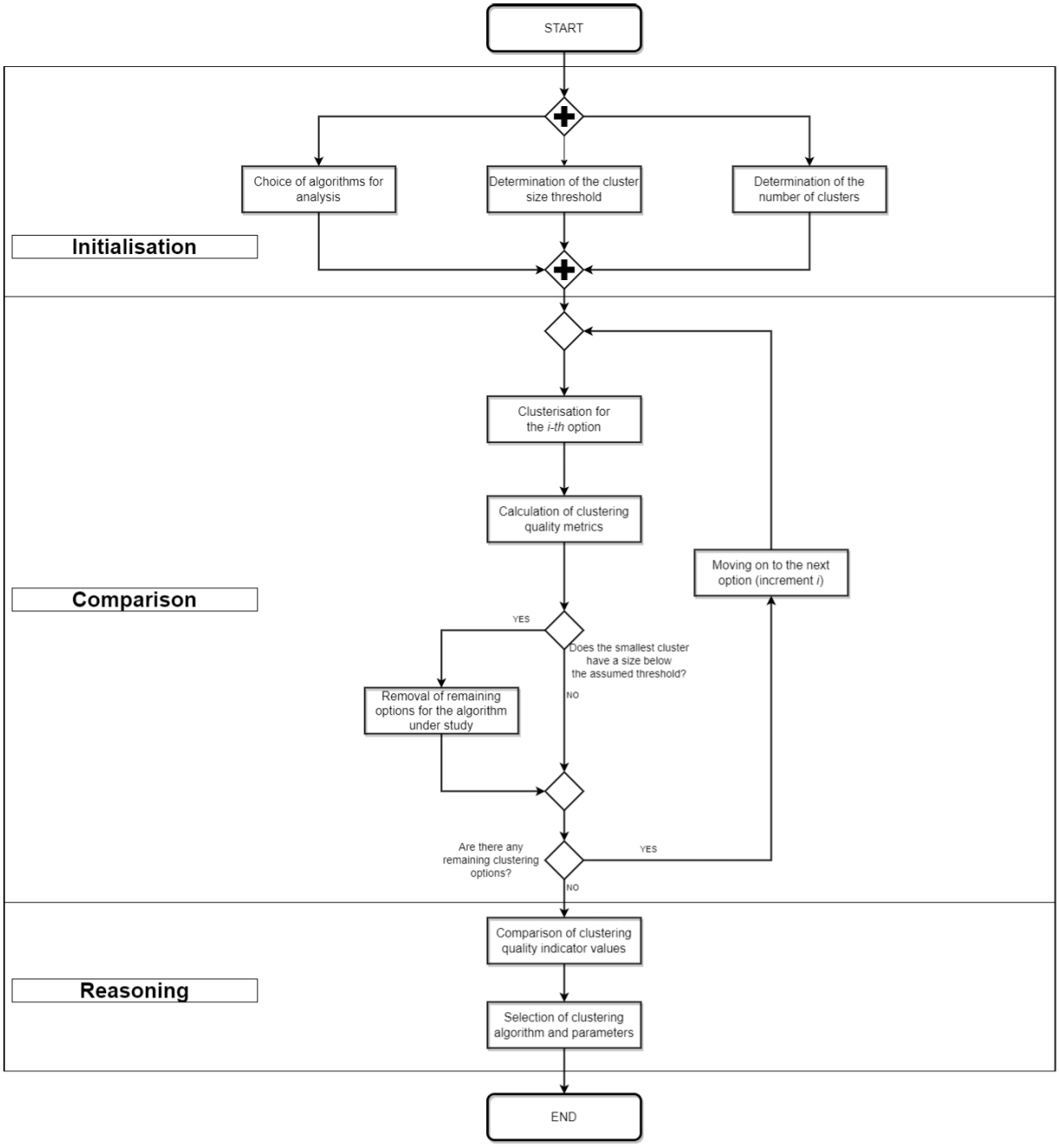


FIGURE 4. Algorithm for selecting clustering method and parameters.

(CH) [48], Dunn Index (DI) [49], and Davies-Bouldin Score (DB) [50]. In addition to these core indicators, composite indicators that aggregate the results of individual measures can also be used [51].

SI measures the similarity of each data point in a cluster to the other data points in the same cluster compared to the nearest neighboring cluster. This measure ranges from -1 to 1, with higher values indicating that the clusters are well separated and have clear boundaries. It should be noted that by adding 1 to this measure, we obtain a measure that is non-negative with higher values indicating a better clustering.

CH measures the ratio of inter-cluster variance to intra-cluster variance. A higher Calinski-Harabasz score indicates that the clusters are better defined and more separated.

DI measures cluster compactness (distance between data points within the same cluster) and cluster separation (distance between different clusters). A higher Dunn Index indicates better cluster quality.

DB takes into account the size (diameter) of the clusters and measures the average similarity of each cluster to its most similar cluster. The lower the value of this metric, the better the clustering method.

For three of these quality indicators, higher values indicate better clustering quality. On the other hand, high values of the DB indicator indicate a poor clustering. However, the Inverted Davies-Bouldin Score (IDB), calculated according to the formula:

$$IDB = \frac{1}{DB} \tag{1}$$

will allow a more consistent interpretation of the results, and in all cases, the clustering methods that achieve the highest values of the metrics will be the best. The calculated values of the clustering quality indicators are used in the reasoning phase.

However, the size of the smallest generated cluster is used as an initial filter. If it is smaller than the set threshold, i.e. the number of online shop customers assigned to a cluster is smaller than the minimum for which it is profitable to prepare a separate variant of the user interface, such an option is rejected and not included in further analysis. At the same time, clustering methods that use the same algorithm with a higher number of clusters (k) as an initialization parameter are removed from the list of potential clustering options. This is due to the fact that as k increases, the minimum size of a cluster is expected to decrease rather than increase.

The final step of this phase is to move on to the next clustering from a prepared list of available options that have not yet been computed or discarded. When all potential clustering options have been considered, the final stage of the proposed algorithm is performed.

3) REASONING

When the values of the indices of clustering quality have been calculated for all the non-rejected options, it is possible to proceed to selecting the best one. To fully automate the decision process, the results obtained for the various indicators should be reduced to a single value that can then be compared between the options.

The simplest solution is to standardize the values of the individual indicators and then calculate an aggregate indicator, taking into account the weights assumed for each component indicator of clustering quality. This approach is called Simple Additive Weighting (SAW), see e.g. [52].

Standardization involves calculating a percentage score for each metric separately. Since we have positive metrics, this can be done most simply by calculating the ratio of the calculated value of an indicator to the maximum value of that indicator in the analyzed set of clustering options, i.e.:

$$X_{ij} = \frac{100 S_{ij}}{S_{max,i}} \tag{2}$$

where: S_{ij} is the measure of the effectiveness of option j according to the $i-th$ indicator of the effectiveness of a clustering, $S_{max,i}$ is the maximum measure of effectiveness over all options according to the $i-th$ indicator of the effectiveness of a clustering and X_{ij} is the standardized score of option j according to the $i-th$ indicator of the effectiveness of the clustering.

It should be noted that the relative difference between the minimum and maximum value of an indicator of the effectiveness of a cluster may depend on the indicator used. For this reason, the following standardized score of the effectiveness of option j according to the $i-th$ measure of the effectiveness of a clustering may be preferred:

$$X_{ij} = \frac{100(S_{ij} - S_{min,i})}{(S_{max,i} - S_{min,i})} \tag{3}$$

where: $S_{min,i}$ is the minimum measure of the effectiveness over all options according to the $i-th$ indicator of the effectiveness of a clustering. In this case, any option associated with the minimum value of a measure of effectiveness is given a standardized score of 0 according to that indicator. Here, for simplicity, we will use Equation 2.

Taking into account the assumed weights for each indicator of clustering quality, the composite score for each of the clustering options (Clustering Option Evaluator - COE), can be derived according to the formula:

$$COE_j = \frac{\sum_{i=1}^{n} w_i X_{ij}}{\sum_{i=1}^{n} w_i} \tag{4}$$

where: COE_j is the composite score of the $j-th$ clustering option and w_i is the weight of the $i-th$ indicator of clustering quality.

In a situation where automatic selection is necessary (e.g. when the platform is delivered on the basis of an SaaS model), the presented algorithm allows a fairly objective evaluation of the possible clustering options and selection of the one assessed to be the best (according to the COE value).

It should be noted that the list of four indicators describing the quality of clustering is not closed, i.e. other metrics can be added to it. On the other hand, it is possible to limit the analysis to a subset of the listed indicators or, if necessary, to only one.

Other methods of multi-criteria decision analysis may also be used, for example TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution). The effectiveness of a solution based on several criteria is assessed according to a standardized measure of distance from an ideal solution (all scores equal to 100 according to Equation 2 and from an ''anti-ideal'' (all scores equal to 0). Compared to the SAW method, given that the average scores of two options are equal, the TOPSIS approach will prefer the option that scores more consistently well (see e.g. [53]).

Using TOPSIS the composite score of an option is given by:

$$COE_j = \frac{100 d_{j,AI}}{d_{j,AI} + d_{j,I}} \tag{5}$$

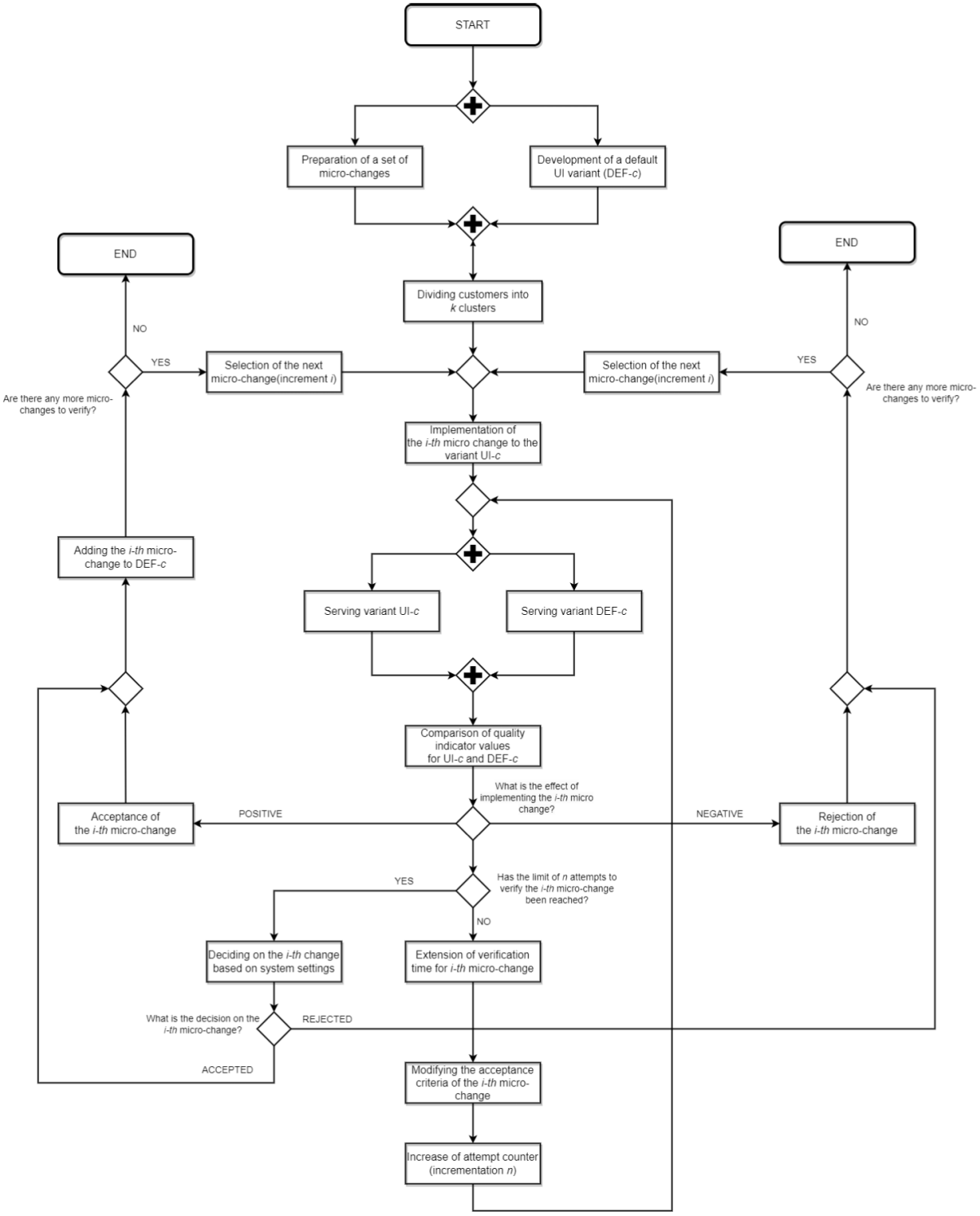


FIGURE 5. Micro-change acceptance algorithm.

where $d_{j,AI}$ is the distance between option j and the anti-ideal and $d_{j,I}$ is the distance between option j and the ideal:

$$d_{j,AI} = \sqrt{\sum_{i=1}^{n} X_{ij}^2}\ ;\ d_{j,AI} = \sqrt{\sum_{i=1}^{n} (100 - X_{ij})^2} \qquad (6)$$

and the scores of options according to each criterion, X_{ij}, can be calculated according to Equation 2. It should be noted that this formula assumes that the criteria have equal weights (by default 100). One may define appropriate weights for the criteria by setting the relative weight of the $j-th$ criterion to

be w_j. The weight of the most important criterion is defined to be 1. In this case,

$$d_{j,AI} = \sqrt{\sum_{i=1}^{n}(w_j X_{ij})^2}\ ;\ d_{j,AI} = \sqrt{\sum_{i=1}^{n}[w_j(100 - X_{ij})]^2} \quad (7)$$

For a comparison of the SAW and TOPSIS procedures see, e.g. [54].

D. ALGORITHM FOR THE ACCEPTANCE OF MICRO-CHANGES

The purpose of the second algorithm (Fig. 5) is to enable the automatic adaptation of user interface variants presented to webshop clients. It is based on the concept of a micro-change - the modification of a specific area of the user interface to improve the user experience for a specific group of customers. In brief, the proposed solution can be described as an automated, sequential A/B testing procedure leading to the development of an optimal UI variant for the groups of e-commerce customers generated using the chosen clustering method. It is a practical implementation of the idea of greater user participation in the processes performed by recommendation systems.

The initialization of the proposed algorithm requires the design and implementation of a collection of i micro-changes, which are then verified in practice. They can be prioritized to determine the order in which the tests are performed, or can be combined into sets containing multiple micro-changes that are tested simultaneously.

In addition to preparing micro-changes, it is necessary to define a benchmark, which is the initial UI variant. This can be defined separately for each cluster and is referred to in the algorithm as DEF-c. It is possible to run the algorithm for the same default variant, but it must be kept in mind that already after the first iteration of self-adaptation, the default UI variants may start to differ between clusters.

In addition, for the described approach to work correctly, it is necessary to determine:

- the maximum number of attempts (n) to verify the impact of a modification in the user interface under analysis,
- the default decision (*Approve*/*Reject*) in case of an ultimately ambiguous result during verification.

The settings of the above parameters can be global, which is useful for deploying the solution in a fully automated form that does not require the support of an administrator or specialist familiar with the nuances of clustering.

The operation of the algorithm starts by dividing the customers of the online store into k clusters. The clustering method and the value of the parameter k are derived from the results of running the algorithm described in section III-C.

The next step is to customize the DEF-c variant by implementing the $i - th$ micro-change from the predefined set. The modified variant will be called UI-c. In the case of parallel UI variant matching for multiple clusters, variants are developed specifically for each cluster, so it is thus possible to consider running the algorithm simultaneously for UI-cx variants, where $x = 1, 2, \ldots, k$.

To reliably assess the impact of the introduced micro-changes, a cluster should then be divided into two groups. One will be presented with the UI-c option and the other with the DEF-c option. The collection of feedback on user behavior and decisions from both groups should last long enough for the results to be considered reliable and relevant. The length of the study can be measured in time (days, weeks), number of sessions, or number of orders placed.

The next operation is crucial for the proposed algorithm. The data collected from both groups of customers should be used to analyze the impact of the micro-changes. To do this, it is necessary to identify indicators whose values will allow further decisions to be made.

From the point of view of e-commerce effectiveness, the conversion rate (CR) and average order value (AOV) are very important metrics:

- CR can be defined as the percentage of customers who placed an order out of the total number of customers who visited an e-shop,
- AOV can be calculated by dividing the total revenue generated from all the orders by the number of orders placed.

In the case of online stores, these two indicators are most often related to orders placed. This means that they refer to macro conversions, which provide final feedback, but do not allow a detailed analysis of customer behavior, in particular the impact of the micro-changes introduced on the actions taken by visitors to the online shop. To fill this gap, an additional indicator, the Partial Conversion Rate (PCR) [55], is proposed:

$$PCR = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{s} CVV_{ij} \quad (8)$$

where: n is the number of sessions related to customers from the analyzed cluster; s is the number of activities within session i, CVV_{ij} is the score of activity j during session i.

Customer Visit Values (*CVVs*) are subjective values assigned to selected actions that a customer can perform in an online store. The choice of specific actions depends on the specifics of the business. If a customer performs an action during the study, it is scored accordingly. If no points have been assigned to an action, its *value* for analysis is 0, but if points have been assigned, they are added to the *value* of the session.

Using this approach, it is possible to identify a number of positive customer behaviors, prioritize them accordingly (by assigning different scores), and then check whether the micro-change implemented has made users more likely to take the action expected by the business owner. The design of this metric provides great flexibility when a detailed study of the impact of the changes made is required. Its calculation does not require the customer to place an order, so less time is required for analysis to obtain reliable results. The proposed PCR indicator is just one way to use the CVV

metric. In addition to the mean, it is also possible to analyze the median, the modal value, the number or percentage of sessions above a certain threshold, etc.

Assuming that the evaluation of the impact of a micro-change is made on the basis of the values of the indicators CR, AOV and PCR (although this list is not closed and can be freely modified), calculated independently for the group of customers presented with the option UI-c and the group presented with the option DEF-c, it is necessary to further define the criteria upon which decisions will be made. In general, the result of this part of the algorithm should be to accept a micro-change (if it has a positive impact on customer behavior), reject it (if it has a negative impact), or postpone the decision (if the impact is unclear).

1) ACCEPTANCE CRITERIA

A micro-change can be accepted if at least one of the conditions **CA1** is met:

(CA1.1):

$$\frac{CR_{UI-c}}{CR_{DEF-c}} > 1 + th \tag{9}$$

(CA1.2):

$$\frac{AOV_{UI-c}}{AOV_{DEF-c}} > 1 + th \tag{10}$$

(CA1.3):

$$\frac{PCR_{UI-c}}{PCR_{DEF-c}} > 1 + th \tag{11}$$

where: th is the acceptance threshold (e.g. 5%).

At the same time, all of the conditions **CA2** should be met:

(CA2.1):

$$\frac{CR_{UI-c}}{CR_{DEF-c}} > 1 - \frac{th}{2} \tag{12}$$

(CA2.2):

$$\frac{AOV_{UI-c}}{AOV_{DEF-c}} > 1 - \frac{th}{2} \tag{13}$$

(CA2.3):

$$\frac{PCR_{UI-c}}{PCR_{DEF-c}} > 1 - \frac{th}{2} \tag{14}$$

If the $i-th$ micro-change is accepted, it is permanently added to the default UI variant for the client cluster under study. The DEF-c variant modified in this way is the basis for verifying further micro-changes in subsequent iterations. The list of modifications awaiting verification is then checked. If there are any left, the next micro-change is taken (the i value is incremented) and the next iteration of the analysis of the impact of UI changes on customer behavior is started. When all micro-changes have been examined, the algorithm ends.

2) REJECTION CRITERIA

In turn, at least one of the conditions **CR1** should be met to reject the micro-change:

(CR1.1):

$$\frac{CR_{UI-c}}{CR_{DEF-c}} < 1 - th \tag{15}$$

(CR1.2):

$$\frac{AOV_{UI-c}}{AOV_{DEF-c}} < 1 - th \tag{16}$$

(CR1.3):

$$\frac{PCR_{UI-c}}{PCR_{DEF-c}} < 1 - th \tag{17}$$

while all conditions **CR2** are met:

(CR2.1):

$$\frac{CR_{UI-c}}{CR_{DEF-c}} < 1 + \frac{th}{2} \tag{18}$$

(CR2.2):

$$\frac{AOV_{UI-c}}{AOV_{DEF-c}} < 1 + \frac{th}{2} \tag{19}$$

(CR2.3):

$$\frac{PCR_{UI-c}}{PCR_{DEF-c}} < 1 + \frac{th}{2} \tag{20}$$

If the $i-th$ micro-change is rejected, then the DEF-c interface variant will not change. The list of changes waiting to be verified is then checked. If any remain, the next micro-changes are taken (the value of i is incremented), and the next iteration of analyzing the impact of UI changes on customer behavior is started. When all micro-modifications have been examined, the algorithm ends.

It should be noted that the conditions for accepting or rejecting a micro-change are clearly disjoint, i.e. it is impossible to both accept and reject a micro-change. However, it is possible that neither condition is satisfied. In such an ambiguous situation, a decision should be postponed to gather more evidence regarding the effectiveness of a micro-change.

3) DECISION POSTPONED

In other cases, the decision is postponed. Further action depends on the number of attempts already made to verify the effect of the implemented micro-change. The maximum number of attempts (n) is set as an input parameter to the proposed algorithm.

If the maximum number of attempts has not yet been reached, the test is continued. First, the analysis of the results will now include new user sessions and all of the previous attempts to verify the effect of the change. Second, the criteria for accepting/rejecting a change can be altered, e.g. by adding or removing an indicator, or by changing (primarily decreasing) the value of the th parameter. Once the new

criteria for analysis have been established, more customer behavior data is collected, with the DEF-c and UI-c options remaining unchanged.

If a micro-change has not been accepted or rejected despite reaching the limit of attempts, a decision is made based on the system settings. Depending on these, the micro-change is either applied to the interface variant or not, and the subsequent actions are analogous to the situation in which a decision was based on a comparison of indicator values.

E. SUMMARY

The key challenge in personalization is ensuring that data-driven customizations actually meet users' fundamental requirements. Our framework addresses this issue by treating user behavior as an implicit expression of their needs and applying a rigorous validation loop to confirm that those needs have been met. The process consists of three stages from the user's perspective:

1) **Inferring Requirements from Behavior** - the initial grouping of users based on their interaction patterns (e.g., navigation paths, feature usage, time spent on the site) is used to group users with similar default requirements. For example, a cluster of users who extensively use filtering options implicitly communicates a requirement for a more powerful and accessible filtering mechanism.
2) **Creating Hypotheses about the User Interface** - the ''micro-changes'' tested in each group are not arbitrary but result from specific design hypotheses aimed at better meeting the inferred requirements. For the previously mentioned cluster, a hypothesis might be that ''making filter options more prominent will reduce the cognitive load and time-to-purchase''. The micro-change is the tangible implementation of this hypothesis.
3) **Verifying that Requirements Are Met Using KPIs** - the system verifies that a requirement has been met by measuring the collective user response using a set of Key Performance Indicators (KPIs). A positive change is interpreted as the user base ''voting'' for the new interface. It provides empirical evidence that the UI change successfully addressed their underlying need, making their task easier, more efficient, or more pleasant. Conversely, a negative or neutral outcome signifies that our hypothesis was incorrect and the proposed solution failed to meet their needs.

Finally, a sustained positive increase in these indicators is considered strong evidence that the user interface change has effectively adapted to the users' goals, thereby meeting their requirements more effectively than the previous interface. This closed-loop system ensures that personalization is driven not by the authors' assumptions, but by the users' own demonstrated needs.

IV. EMPIRICAL RESEARCH

Examples of the use of the proposed algorithms are presented on the basis of an experimental study conducted using the AIM^2 platform, developed by the e-commerce vendor Fast White Cat. This system contains all the components necessary to collect and analyze data on customer behavior and, on this basis, to design and deploy dedicated UI variants [55]. The algorithms described in this paper have been implemented within the platform. The results presented below illustrate the decision-making process for one of the experiments conducted during the solution validation phase. This is an illustration of the implementation of the concept of user-centric personalization. In the approach described, UI variants are guided by actual customer behavior and decisions, allowing them to be improved through a feedback loop.

A. SELECTION OF THE CLUSTERING METHOD

For the selection of the clustering method, customer behavior data was collected from an online clothing store and consisted of 438,261 user sessions. This was a learning dataset that was clustered using several different algorithms. Detailed logs include additional information about the device used, country and language, and previous visits. The most important part of the data was the type of pages loaded during the users' activity, taking into account characteristics such as the category of the object viewed, the time spent, or whether the activity ended with a purchase. In addition, the data set was enriched with information about existing products and their categories.

The clustering methods used and the parameters regarding the size and number of clusters are described below:

- algorithms analyzed: Agglomerative [56], BIRCH [57], DBSCAN [58], K-means [59], [60], K-medians [61],
- minimum cluster size: 5% of the total customer population,
- expected number of clusters: 4.

The indicators SI, CH and IDB were used as a basis for the analysis of the clustering quality. Each indicator was given the same weight of 33.3(3)%.

The values of the indicators relevant to the algorithm under study are shown in Tab. 1.

If the business context and the requirements arising from the design of the specific UI options are not taken into account, the proposed algorithm could also be used to compare the five grouping options considered. On the basis of Equation 2, the values of the standardized ratios used to compute the aggregate score of each clustering method based on both SAW and TOPSIS are shown in Tab. 2. It should be noted that, in practice, it is more efficient to first eliminate clustering methods that do not satisfy the constraints resulting from the context (i.e. regarding the number and size of clusters), but a fuller analysis is given here for illustration. According to both SAW and TOPSIS, K-means clustering has the highest composite score and would be selected as the best

TABLE 1. Indicator values for the clustering options analysed.

Metric	UoM	Agglomerative	BIRCH	DBSCAN	K-means	K-medians
Duration	sec.	2071	779	1003	564	1043
No. of clusters		4	4	1148	4	4
Min. cluster size	users	17904	6182	3	11789	763
Min. cluster size	%	19.2597	6.6501	0.0032	19.1360	0.8208
Max. cluster size	users	28387	45246	35133	28504	49234
Max. cluster size	%	30.5365	48.6720	37.7933	30.6623	52.9620
SI		0.1636	0.0762	0.2678	0.1710	0.2150
CH		11675.8058	6152.0925	69.9466	12590.2920	7000.9324
IDB		0.3588	0.3086	0.8235	0.4075	0.3791

TABLE 2. Assessment of the clustering options in a context-free setting.

Metric	Agglomerative	BIRCH	DBSCAN	K-means	K-medians
sSI	61.0974%	28.4538%	100.0000%	63.8447%	80.2974%
sCH	92.7366%	48.8638%	0.5556%	100.0000%	55.6058%
sIDB	43.5737%	37.4739%	100.0000%	49.4787%	46.0370%
SAW	65.8026%	38.2638%	66.8519%	**71.1078%**	60.6467%
TOPSIS	63.3823%	38.5998%	58.7140%	**67.4176%**	59.7950%

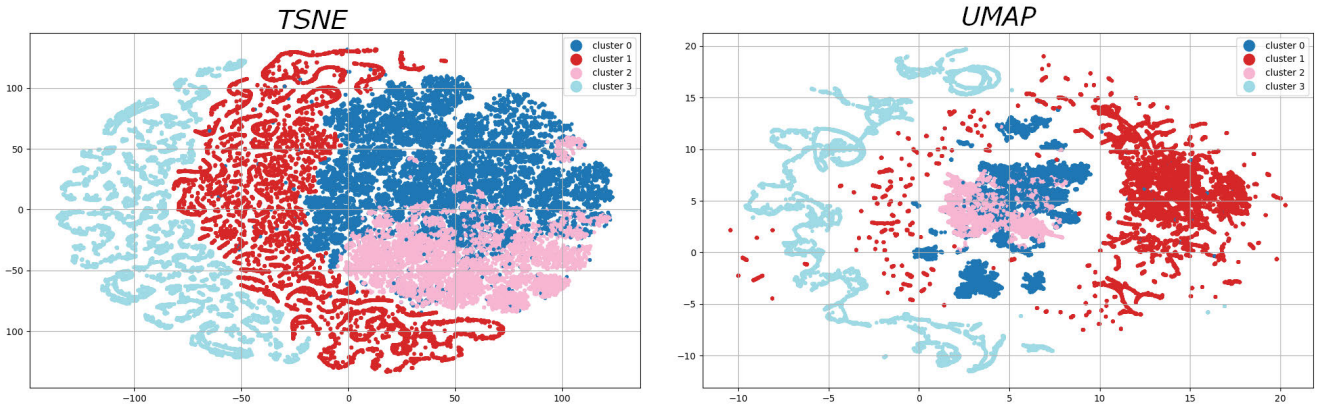


FIGURE 6. Visualisation of recommended clustering.

TABLE 3. Assessment of the clustering options after discarding DBSCAN.

Metric	Agglomerative	BIRCH	K-means	K-medians
sSI	76.0889%	35.4355%	79.5103%	100.0000%
sCH	92.7366%	48.8638%	100.0000%	55.6058%
sIDB	88.0654%	75.7374%	100.0000%	93.0440%
SAW	85.6303%	53.3456%	**93.1701%**	82.8833%
TOPSIS	84.3101%	53.0066%	**88.7869%**	76.6460%

TABLE 4. Assessment of the clustering options after discarding DBSCAN and K-medians.

Metric	Agglomerative	BIRCH	K-means
sSI	95.6969%	44.5672%	100.0000%
sCH	92.7366%	48.8638%	100.0000%
sIDB	88.0654%	75.7374%	100.0000%
SAW	92.1663%	56.3895%	**100.0000%**
TOPSIS	91.6152%	55.9312%	**100.0000%**

option based on the proposed algorithm when the business context is taken into account.

In practice, the business requirements cannot be overlooked, as this could make it impossible to achieve the main objective. According to the proposed algorithm, the business requirements are addressed in two aspects. The first is the predefined number of clusters. Many methods explicitly require this value as an initialization parameter, but the DBSCAN algorithm does not. In some cases this can be a great advantage, but if the purpose of clustering is to prepare groups of clients to be served by a dedicated interface, it is a disqualifying disadvantage. In the case analyzed, the DBSCAN method created 1148 clusters, the smallest of which had only three clients. For obvious reasons, this is not a commercially acceptable result. This means that this algorithm should be excluded from further analysis. It should also be noted that the performance of DBSCAN according to the three criteria is very inconsistent. As a result, although it is rated second according to the SAW procedure, it is only rated fourth according to the TOPSIS procedure.

The updated values of the indicator, after DBSCAN has been discarded, are shown in Tab. 3. In this case, the SAW and TOPSIS methods rank the clustering methods in the same order. Again, the highest composite score was calculated for K-means, confirming the high quality of this clustering.

The second business requirement was the size of the smallest cluster. According to the proposed algorithm,

options should be rejected when the smallest cluster contains less than the threshold number of customers. The study assumed that a single cluster should contain at least 5% of the total number of clustered customers. In the case analyzed, this condition was not met by K-medians clustering, where the smallest cluster contained only 0.82% of all the users. Therefore, this clustering method should also be rejected, and the updated indicator values are shown in Tab. 4. Finally, the algorithm described in this example would compare 3 clustering options: Agglomerative, BIRCH, and K-means. The computed values of the aggregated scores indicate that K-means is the best method and should be selected for further use within the AIM^2 platform. These clusters are illustrated using the following two approaches: t-Distributed Stochastic Neighbor Embedding (TSNE) and Uniform Manifold Approximation and Projection (UMAP) (see Fig. 6).

It should be noted that such a recommendation applies only to a specific learning dataset and is not directly transferable to other implementations. Therefore, the process of selecting the optimal clustering approach should always be performed for each set of data from different webshops. In addition, it is worth mentioning that automated methods, including those from the Multiple Criteria Decision Analysis (MCDA) group [62], can be used to select a clustering technique, allowing the parameterization of the clustering process to be approached in a formal way. Admittedly, in the case of this algorithm, users are not central to the mechanism, but their behavior determines how clustering occurs. Therefore, it is possible to dynamically adapt the clustering rules to customers, which is an innovative approach compared to existing market solutions that impose segmentation rules and do not dynamically adapt them to the specifics of the users of a particular e-commerce platform.

B. ACCEPTANCE/REJECTION OF A MICRO-CHANGE

The second algorithm described makes it possible to analyze the impact of micro-changes made to the user interface of an online shop. To demonstrate its performance, the case of verifying a set of modifications presented to four groups of customers was chosen. These were generated during the study described in section IV-A and are derived from the application of the K-means algorithm. This is an example of a mechanism whose central element is the user. Their decisions and behaviors determine the direction, scope, and depth of UI changes.

The study was performed for all of the generated customer clusters, i.e. the proposed algorithm was applied in parallel for each of the various user groups. In all four cases, a standard webshop layout was adopted as the initial default user interface and a modification was proposed for verification, consisting of a set of 12 modifications covering the web pages: home page, listing and product card. The same modifications were analyzed in all customer groups.

The online shop customers (each ascribed with a unique identifier) were divided into four ($k = 4$) clusters of 28,504, 27,871, 17,789 and 18,797 users. It was also assumed that the number of iterations in the decision process would be limited to one ($n = 1$) and that if the recommendation was ambiguous, then the change would be rejected. To determine whether an impact is significant, we established a practical significance threshold of 6% ($th = 6\%$). This business-defined threshold ensures that only changes inducing a meaningful and tangible impact on key metrics are accepted. While statistical significance testing (e.g., t-tests) confirms whether an observed effect is unlikely to be due to random chance, the concept of practical significance used here answers a different, but crucial, business question: is the magnitude of the change large enough to justify its implementation? This pragmatic approach prioritizes the business value of UI modifications.

For the purposes of the study, each customer group was further split in half. One half was presented with a custom UI and the other half was presented with a standard UI. User behavior data were collected for an additional 27 days, resulting in 99193 sessions in the online store.

1) PARAMETRIZATION OF THE STUDY

Three indicators were used to verify the impact of the changes studied:

- CR, calculated as the ratio of the number of orders placed to the number of user sessions in a specific group,
- AOV, calculated as the average order value to be paid,
- PCR, calculated for the following actions (points awarded in brackets):
  - -- switching from the homepage to the listing (10),
  - -- switching from the home page to the product page (10),
  - -- switching from the listing to the product card (10),
  - -- switching between product pages (5),
  - -- adding a product to the cart from the product card or the listing (20).

The indicator values for each customer cluster are shown in Tab. 5.

2) APPLICATION OF THE ALGORITHMS

For *Cluster*1:

1) two acceptance conditions CA1 are met (CA1.1 and CA1.3), but at the same time one of the conditions CA2 is not met (CA2.2), so the change cannot be accepted,
2) none of the rejection conditions CR1 are met, so the change cannot be rejected.

This means that, according to the algorithm, the verification did not result in a clear decision about the considered modification.

In such a situation, the algorithm provides for the possibility of a further iteration of the study, but since $n = 1$ was assumed, this option was not used.

TABLE 5. Results of the study on the impact of UI changes.

Cluster	Sample	UI variant	*CR*	*CR* diff.	*AOV*	*AOV* diff	*PCR*	*PCR* diff.
1	657	UI-*c*	3.04%	+72.73%	40.02	-4.03%	30.60	+14.91%
	682	DEF-*c*	1.76%		41.70		26.63	
2	1851	UI-*c*	1.62%	+27.56%	47.67	+68,09%	30.54	+6.56%
	1650	DEF-*c*	1.27%		28.36		28.66	
3	3898	UI-*c*	3.13%	+2.96%	43.81	+1,55%	37.29	-18.24%
	3683	DEF-*c*	3.04%		43.14		45.61	
4	3275	UI-*c*	3.15%	+18.87%	43.02	-0.09%	45.61	+7.12%
	3317	DEF-*c*	2.65%		43.06		42.58	

According to the assumption made about how to deal with ambiguity in the results of the analysis, the change in question would be **rejected**.

For *Cluster*2:

1) the three acceptance conditions CA1 are met, and all of the conditions CA2 are met, so the change would be **accepted**.

For *Cluster*3:

1) none of the conditions CA1 for acceptance are met,
2) one of the conditions for rejection of CR1 is met (CR1.3),
3) all of the conditions for rejecting CR2 are met, so there is reason to **reject** the change.

For *Cluster*4:

1) two acceptance conditions CA1 are met (CA1.1 and CA1.3), and all of the conditions CA2 are met, so the change would be **accepted**.

The result of the algorithm would be the addition of the confirmed changes to the default UI variants (DEF-*c*) for clusters 2 and 4. For clusters 1 and 3, the default UI variant would remain unchanged.

The next step for the automatic operation of the UI self-adaptation mechanism would be to verify the next micro-change from the prepared set, taking into account the changes introduced as a result of the described procedure. The end result is UI variants that have evolved based on user decisions and behavior, thanks to an innovative approach to the role of the customer in the e-commerce personalization process. The proposed solution combines user involvement, gives the user indirect control over the direction of changes, and thus helps to better design and tailor the UI.

V. LIMITATIONS AND FUTURE RESEARCH

Despite promising results, our structure has several limitations that suggest directions for future research. The primary limitation, as highlighted by our empirical design, is the focus on a single e-commerce platform within the online apparel domain. This raises questions about the generalizability of our findings to other business contexts.

However, the core methodological framework - leveraging unsupervised clustering to identify behavioral segments and using a closed-loop, KPI-driven mechanism to validate UI micro-changes - is inherently domain-agnostic. The fundamental principles can be applied across different e-commerce sectors, although the specific inputs and outputs would vary, e.g.:

- in consumer electronics user segments might cluster around ''spec-comparison shoppers'' versus ''brand-loyal upgraders'' and the UI micro-changes tested could involve personalizing product comparison tools, filtering mechanisms for technical specifications, he presentation of customer reviews, etc.;
- in online groceries our framework could identify segments like ''weekly stock-up shoppers'' versus ''recipe-based explorers'' and UI adaptations might involve personalizing quick re-order functionalities for the former and promoting ingredient bundles for the latter.

Therefore, while the specific clusters and successful UI changes identified in this study are context-dependent, the adaptive framework itself provides a robust methodology for personalization in other domains. Validating this cross-domain applicability through further large-scale empirical studies remains a key direction for future research. The auto-adaptation process can be time-consuming, as collecting sufficient data for low-activity clusters can take a long time. A significant challenge is the ''cold start'' problem: our system cannot personalize the experience for new users. In the future, it would be possible to explore the integration of a default user interface for new users, based on the most effective variant from a similar, earlier cluster. In addition, privacy issues require continuous improvement; future research could focus on federated learning or other privacy-enhancing technologies to further minimize data exposure. Finally, optimizing the selection and granularity of ''micro-changes'' to balance impact with analytical transparency remains an important area for further research.

It should be noted that our evaluation framework prioritizes practical significance by using a predefined KPI threshold to ensure business-relevant outcomes. A valuable extension for future research or industrial-scale deployment would be to integrate formal statistical significance testing into the A/B testing phase. This would add an additional layer of rigor by systematically verifying that the observed performance uplifts are not only practically meaningful but also statistically robust, thereby reducing the risk of implementing changes based on random fluctuations in user behavior.

Additionally, while this study validates the effectiveness of our closed-loop, KPI-driven framework, it does not include a direct comparative analysis against alternative personalization models, such as those employing reinforcement learning for UI optimization. Future research should aim to benchmark the performance and computational efficiency of our sequential A/B testing approach against these other emerging methods to better situate its practical advantages and disadvantages.

In summary, the use of multi-variant user interfaces in e-commerce is a further step towards tailoring communication to the needs, behaviors and decisions of individual customers. It thus fits into the general trend of personalizing UX and the concept of involving users more fully in the recommendation system. The proposed algorithms can form a basis for the development of solutions that go beyond the standard model of a one-size-fits-all UI in online shops, thereby increasing customer satisfaction, which can translate into tangible benefits to a business. The implementation of the proposed solution also enables the automation of decision-making processes, shifting the burden of decision making from humans to the recommendation system. Nevertheless, due to the evolutionary nature of the proposed algorithms, they still have weaknesses. Precise identification of these weaknesses will direct their potential development and optimization.

Furthermore, the implementation of any adaptive system carries ethical and user experience considerations that warrant careful attention. There is a potential risk that deep personalization could create ''filter bubbles'', narrowing a user's exposure to new or diverse products. The proposed framework mitigates this by focusing on adapting the UI structure and interaction mechanisms rather than strictly filtering content. Another risk is user confusion or frustration resulting from an interface that changes too frequently. Our approach addresses this by making changes evolutionary and validating them at the cluster level, not for each individual, which promotes a more stable experience. Future work should continue to explore mechanisms to ensure fairness, transparency, and user control, preventing the system from developing into a tool for manipulation and instead ensuring it remains a user-centric tool for reducing cognitive load and improving task efficiency.

VI. CONCLUSION

The practical implementation of a self-adjusting e-commerce interface mechanism has enormous potential to improve user comfort and increase business success. This study presents and verifies an innovative structure based on two algorithms for automating this process. The results of the study and their implications are summarized below.

A. INTERPRETATION OF KEY FINDINGS

The results of our experiments show that adapting the user interface to the hidden needs of different user segments can bring significant benefits. The success of the K-means algorithm, selected by our context-aware selection method, confirmed that there are distinct and behaviorally consistent user groups within a typical e-commerce audience. Subsequent verification of micro-changes revealed a strong cause-and-effect relationship between the customized user interface design and user behavior.

For example, the high 68.09% increase in average order value in Cluster 2 suggests that the UI modifications, which likely focused on enhancing product discovery and cross-selling opportunities, directly corresponded to the exploratory nature of this segment. These users responded positively to an interface that encouraged browsing and discovering new items.

Conversely, the rejection of these same changes for Cluster 3 is an equally important finding. This outcome strongly suggests that Cluster 3 represents a different user persona. For this segment, the changes that benefited the users in Cluster 2 may have introduced unwanted friction, such as additional clicks, visual clutter, or a disrupted navigational path. This hypothesis is supported by the significant negative impact on the Partial Conversion Rate, which measures the efficiency of micro-conversions within the purchase funnel. The negative result for Cluster 3 is not a failure of the personalization effort but highlights the validation process. It demonstrates that our KPI-based verification effectively prevents the implementation of ineffective or harmful user interface modifications, acting as a key safety mechanism to protect the user experience for all segments.

B. THEORETICAL IMPLICATIONS

From a theoretical point of view, these studies make two main contributions. First, they propose a complete, closed-loop model of self-adjustment that extends personalization beyond content recommendations to the evolution of the user interface. It provides a practical roadmap for a system in which users are not merely recipients of personalization but active drivers of interface design through a continuous feedback loop. Second, it contributes a contextual quality framework for selecting clustering algorithms that formally integrates technical performance metrics with practical business constraints, offering a more robust approach than relying solely on technical metrics.

C. PRACTICAL IMPLICATIONS

For e-commerce practitioners, this work can provide a concrete framework for moving beyond static, one-size-fits-all interfaces. The main practical takeaway is the potential of an automated, data-driven tool that can continuously optimize user experiences for different customer segments, leading to significantly higher revenue and engagement. Our results provide strong evidence that investing in user interface personalization can yield a direct and significant return on investment. The PCR metric we introduced also provides a flexible new metric that allows practitioners to measure engagement beyond simple, well-known, and widely used macro-conversion metrics.

DATA AVAILABILITY

The datasets used in the study described in the paper are available for download from https://doi.org/10.7910/DVN/QGB45G

DECLARATION OF GENERATIVE AI IN SCIENTIFIC WRITING

AI-based tools were used for language correction in parts of the article.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

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